JAVA PROGRAM REPORT

Student Performance Evaluation

Programming Language:	Java
Topic:	User Input, Arithmetic and Conditional Statements
Program File:	StudentPerformance.java

1. AIM

To write a Java program that reads a student's name and marks in three subjects, calculates the total and average, and assigns a performance category based on the average.

2. DESCRIPTION

The program uses Scanner to accept the student's name and three integer marks. It calculates total marks and average marks, then uses an if-else-if ladder to classify performance as Excellent, Good, Average, or Fail.

3. ALGORITHM

1. Create a Scanner object.
2. Read the student name and marks for three subjects.
3. Calculate total = m1 + m2 + m3.
4. Calculate average = total / 3.0.
5. If average ≥ 75, display Excellent.
6. Else if average ≥ 60, display Good.
7. Else if average ≥ 40, display Average; otherwise display Fail.
8. Close the Scanner and stop.

4. SOURCE CODE

Save the program as **StudentPerformance.java**, matching the public class name.

```
import java.util.Scanner;

public class StudentPerformance {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Marks in Subject 1: ");
        int m1 = sc.nextInt();

        System.out.print("Enter Marks in Subject 2: ");
        int m2 = sc.nextInt();

        System.out.print("Enter Marks in Subject 3: ");
        int m3 = sc.nextInt();

        int total = m1 + m2 + m3;
        double average = total / 3.0;

        System.out.println("\n--- Student Performance ---");
        System.out.println("Student Name: " + name);
        System.out.println("Total Marks: " + total);
        System.out.println("Average Marks: " + average);

        if (average >= 75) {
            System.out.println("Performance: Excellent");
        }
        else if (average >= 60) {
            System.out.println("Performance: Good");
        }
        else if (average >= 40) {
            System.out.println("Performance: Average");
        }
        else {
            System.out.println("Performance: Fail");
        }

        sc.close();
    }
}
```

5. PROGRAM EXPLANATION

1. Scanner: Reads the student's name and subject marks from keyboard input.

2. Total: Adds the three subject marks together.

3. Average: Divides total by 3.0 so the result can include decimal values.

4. if-else-if ladder: Selects the performance category using the average thresholds.

5. sc.close(): Closes the Scanner after all input is processed.

6. EXECUTION AND OUTPUT

Compile and run:

```
C:\Java> javac StudentPerformance.java
C:\Java> java StudentPerformance
```

Test Case 1: Excellent performance

```
Enter Student Name: Ravi
Enter Marks in Subject 1: 90
Enter Marks in Subject 2: 80
Enter Marks in Subject 3: 85
--- Student Performance ---
Student Name: Ravi
Total Marks: 255
Average Marks: 85.0
Performance: Excellent
```

6. EXECUTION AND OUTPUT (CONTINUED)

Test Case 2: Good performance

```
Student Name: Meena
Total Marks: 195
Average Marks: 65.0
Performance: Good
```

Test Case 3: Average performance

```
Student Name: Arun
Total Marks: 135
Average Marks: 45.0
Performance: Average
```

Test Case 4: Fail

```
Student Name: Kiran
Total Marks: 75
Average Marks: 25.0
Performance: Fail
```

7. TEST CASE SUMMARY

Test Case	Marks	Average	Expected Result	Status
1	90, 80, 85	85.0	Excellent	PASS
2	65, 60, 70	65.0	Good	PASS
3	45, 40, 50	45.0	Average	PASS
4	20, 30, 25	25.0	Fail	PASS

8. RESULT

The program compiled and ran successfully. It correctly calculated totals and averages and displayed the expected performance category for all four test cases.

9. CONCLUSION

This program demonstrates Java keyboard input, arithmetic calculations, and conditional decision-making to evaluate student performance from three subject marks.